

PRIYANKA S

Cloud Engineer | AWS & DevOps

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Reduced manual settlement-validation effort by ~59% FTE and cut average case handling time from 11.25 to 7.5 minutes through automated AWS pipelines for a major UK energy supplier.

SUMMARY

Cloud Engineer with 1.5+ years of hands-on AWS experience designing and delivering production-grade serverless automation for a major UK energy supplier. AWS Certified Solutions Architect – Associate (Aug 2026). Skilled in Infrastructure as Code (Terraform, CloudFormation), multi-account and multi-region architecture, serverless compute (Lambda, SQS, S3), hybrid networking (VPC, Transit Gateway, Route 53 Resolver), and CI/CD automation with GitHub Actions and keyless OIDC. Also bring applied AI/ML experience (deep learning, NLP) from prior work, adding a broader technical toolkit to cloud and platform engineering roles.

CORE SKILLS

Cloud (AWS)	Lambda, SQS, S3, DynamoDB, ECR, API Gateway, Cognito, Secrets Manager, KMS, VPC, Transit Gateway, Route 53 Resolver, VPC Endpoints, CloudWatch, WAF, SNS
DevOps & IaC	Terraform, CloudFormation, Docker, GitHub Actions, OIDC, remote state with S3 and DynamoDB locking
Languages	Python, SQL
Data & BI	Oracle, Snowflake, Power BI
AI / ML	Deep Learning, NLP, Cohere, pandas, NumPy, Streamlit

PROFESSIONAL EXPERIENCE

Tata Consultancy Services (TCS) - AI/ML Engineer — Cloud Infrastructure & DevOps Mar 2024 - Present
Cloud automation for a major UK energy supplier (smart metering and settlement)

Settlement Validation Automation Pipeline (SR0074)

- Designed a **4-Lambda fan-out architecture**: S3 triggers a split Lambda that batches MPAN files into groups of 50; a prefetch Lambda caches Snowflake data once per job; worker Lambdas process batches in parallel against Oracle; and a merge Lambda creates the final Excel report.
- Built the solution with **Terraform**, including private subnets, Transit Gateway connectivity to on-prem Oracle, Route 53 Resolver, KMS, Secrets Manager, SQS with DLQ, VPC endpoints, and WAF.
- Implemented **multi-account, multi-region active/DR deployment** with isolated Terraform state per environment and central control-plane governance.
- Implemented **GitHub Actions CI/CD with keyless OIDC**, running terraform plan on pull requests and gated apply on merge.
- Packaged the runtime as a **Docker image with Oracle Instant Client and Python**, published to Amazon ECR.
- Validated the solution on **live production data**, reducing average case handling time from **11.25 to 7.5 minutes** and providing an estimated **~10.75 FTE** capacity benefit; presented the business case and secured go-ahead for production rollout.

DC Validation ETL Pipeline (MPAN Data)

- Delivered an **AWS Lambda ETL pipeline** that ingested Excel MPAN files from S3, validated records in parallel using 50 concurrent workers, connected to Oracle with service fallback, and loaded results to Snowflake.
- Ran within a **private, no-internet VPC** and managed production and test environments with Terraform, S3 and DynamoDB state locking, and GitHub Actions.
- Reduced the process's FTE requirement from **29.72 to 12.18**, a **~59% reduction**, by automating validation and cross-referencing previously performed manually.
- Established a **reusable blueprint** that informed the later SR0074 settlement-validation rollout.
- Championed **security best practices**, including least-privilege IAM, encrypted state and secrets, and automated IaC scanning in CI/CD.

PROJECTS

- Serverless E-Commerce Platform (AWS)** - Built a cloud-native storefront with Cognito authentication, API Gateway, Lambda, DynamoDB, and SQS/SNS event-driven order processing, deployed through automated CI/CD and OIDC-based AWS access.
- Cloud Security Monitor / SIEM (AWS)** - Developed real-time AWS monitoring using CloudTrail and EventBridge, with a Lambda classifier that assigns severity and triggers SNS alerts for critical events.
- Mental Health Wellbeing Chatbot (AI/ML)** - Designed an NLP-based assistant using Cohere and Streamlit to provide empathetic, context-aware wellbeing support.
- Predictive Analytics & BI** - Designed interactive Power BI dashboards with forecasting models, improving decision-making efficiency by 30%.

EDUCATION & CERTIFICATIONS

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| B.Tech, Computer Science & Engineering - Mar Baselios Christian College of Engineering | GPA: 9.1 / 10 |
| AWS Certified Solutions Architect - Associate (SAA-C03) | Aug 2026 |